

- Which of the following liquids will have the highest freezing point?
 - a. aqueous FeI_3 (0.030 m)
 - b. aqueous $\text{C}_6\text{H}_{12}\text{O}_6$ (0.050 m)
 - c. aqueous COI_2 (0.030 m)
 - d. aqueous NaI (0.030 m)
 - e. pure H_2O
- When one mole of benzene is vaporized at a constant pressure of 1.00 atm and at its boiling point of 353.0 K, 30.79 kJ of energy (heat) is released and the volume change is +28.90 L. What is ΔE for this process? (1 L-atm = 101.3 J)
 - a. -33.72 kJ
 - b. -27.86 kJ
 - c. 27.86 kJ
 - d. 33.72 kJ
- Which probably has the lowest boiling point at 1.00 atm?
 - a. H_2Te
 - b. H_2S
 - c. H_2Se
 - d. H_2O
- Given the following data, determine the rate law for the reaction: $\text{X} + \text{Y} \rightarrow 2\text{Z}$

Exp	[X] (M)	[Y] (M)	Rate (M/s)
1	0.0300	0.0100	3.4×10^{-4}
2	0.0150	0.0100	8.5×10^{-5}
3	0.0150	0.0400	3.4×10^{-4}

 - a. Rate = $k[\text{X}]^2[\text{Y}]$
 - b. Rate = $k[\text{Y}]$
 - c. Rate = $k[\text{X}]^2[\text{Y}]^2$
 - d. Rate = $k[\text{Y}]^2$
- Which of the following liquids will have the lowest freezing point?
 - a. aqueous COI_2 (0.030 m)

- b. pure H_2O
 - c. aqueous $\text{C}_6\text{H}_{12}\text{O}_6$ (0.60 m)
 - d. aqueous FeI_3 (0.24 m)
 - e. aqueous KF (0.50 m)
6. Which of the following choices has the compounds correctly arranged in order of increasing solubility in water? (least soluble to most soluble)
- a. $\text{CCl}_4 < \text{CHCl}_3 < \text{CH}_3\text{OH}$
 - b. $\text{CHCl}_3 < \text{CCl}_4 < \text{NaNO}_3$
 - c. $\text{CH}_3\text{OH} < \text{CH}_4 < \text{LiF}$
 - d. $\text{CH}_4 < \text{NaNO}_3 < \text{CHCl}_3$
7. $\text{A} \rightarrow \text{B}$ is a first-order reaction. What is the half-life (in seconds) for the reaction?
- | Time (sec) | [A] (M) |
|------------|---------|
| 0.0 | 1.22 |
| 5.0 | 0.86 |
| 10.0 | 0.61 |
| 15.0 | 0.43 |
| 20.0 | 0.31 |
- a. 20.0
 - b. 10.0
 - c. 15.0
 - d. 5.0
8. When one mole of benzene is vaporized at a constant pressure of 1.00 atm and at its boiling point of 353.0 K, 30.79 kJ of energy (heat) is absorbed and the volume change is -28.90 L. What is ΔE for this process? (1 L-atm = 101.3 J)
- a. 33.72 kJ
 - b. -27.86 kJ
 - c. 27.86 kJ
 - d. -33.72 kJ
9. A liquid placed in a capillary tube has the meniscus shown below. The types of forces that create this situation are:

- a. cohesive and adhesive forces have no effect
- b. Adhesive forces are stronger than cohesive forces
- c. Cohesive forces are stronger than adhesive forces

10. If the rate of appearance of O_2 in the reaction: $2O_3(g) \rightarrow 3O_2(g)$ is 0.250 M/s over the first 16.50 s, how much oxygen will form during this time?

- a. 3.38 M
- b. 4.13 M
- c. 0.25 M
- d. 2.63 M
- e. 1.38 M

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
E	A	B	A	D	A	B	B	B	A

Questions (Continued):

11. When a 14.46 g sample of $KClO_3$ (122.5 g/mol) is dissolved in 200. g of water in a coffee-cup calorimeter, the temperature falls from 25.0 °C to 18.4 °C. Assume an isolated system where the specific heat of the solution is 4.184 J/g °C. What is the enthalpy change (ΔH , in kJ/mol) for the solution process of $KClO_3$ dissolving in water?

- a. 36.4 kJ/mol
- b. 44.5 kJ/mol
- c. 50.2 kJ/mol
- d. 40.0 kJ/mol

12. What is the molarity of an aqueous solution labeled "7.3% glucose ($C_6H_{12}O_6$, 180 g/mol) by weight?

- a. 0.44 m
- b. 0.67 m
- c. 0.52 m
- d. 0.34 m

- e. 0.26 m

13. What types of intermolecular forces exist between CH_3OH and H_2O ?

- a. dispersion forces, dipole-dipole, and hydrogen bonding
- b. dispersion forces and ion-dipole
- c. dispersion forces, dipole-dipole, and ion-dipole
- d. dipole-dipole and ion-dipole
- e. dispersion forces, hydrogen bonding, dipole-dipole, and ion-dipole

14. A 5.00 mL sample of solution has 3.6×10^{-4} g of Ca^{2+} . Calculate the concentration of Ca^{2+} in ppm? The density of the solution is 1.0 g/mL.

- a. 38 ppm
- b. 56 ppm
- c. 84 ppm
- d. 72 ppm

15. A 250 mL solution containing 14.5 g of a polymer in toluene had an osmotic pressure of 0.055 atm at 27 °C. What is the molar mass of the polymer?

- a. 26000 g/mol
- b. 38000 g/mol
- c. 32000 g/mol
- d. 15000 g/mol
- e. 18000 g/mol

16. Complete combustion of 1.00 mol of methane (CH_4) liberates 890.3 kJ of heat. Given that $\Delta H^\circ_f(\text{CO}_2) = -393.5$ kJ/mol and $\Delta H^\circ_f(\text{H}_2\text{O}) = -285.8$ kJ/mol, calculate the standard enthalpy of formation of methane.

- a. 74.8 kJ/mol
- b. -1855 kJ/mol
- c. -74.8 kJ/mol
- d. 1855 kJ/mol

17. The fluorocarbon compound $\text{C}_2\text{Cl}_3\text{F}_3$ (187.4 g/mol) has a normal boiling point of 47.6 °C. The specific heats of $\text{C}_2\text{Cl}_3\text{F}_3$ (l) and $\text{C}_2\text{Cl}_3\text{F}_3$ (g) are 0.91 J/g K and 0.67 J/g K, respectively. The heat of vaporization for the compound is 27.49 kJ/mol. Calculate the heat required to convert 54.0 g of $\text{C}_2\text{Cl}_3\text{F}_3$ from a liquid at 10.85 °C to a gas at 87.40 °C.

- a. 15.30 kJ
- b. 9.10 kJ
- c. 13.24 kJ
- d. 11.17 kJ

18. For the reaction diagram shown, which of the following statements is false?

- a. For the forward reaction, line W represents the activation energy, and the forward reaction would be exothermic by the quantity Y.
- b. The reverse reaction should be faster than the forward reaction.
- c. For the reverse reaction, line X represents the activation energy, and the reverse reaction would be endothermic by the quantity Z.
- d. In both the forward and the reverse direction, point B represents the Transition State.

19. When a 16.46 g sample of KClO_3 (122.5 g/mol) is dissolved in 200. g of water in a coffee-cup calorimeter, the temperature falls from 25.0 °C to 18.4 °C. Assume an isolated system where the specific heat of the solution is 4.184 J/g °C. What is the enthalpy change (ΔH , in kJ/mol) for the solution process of KClO_3 dissolving in water?

- a. 44.5 kJ/mol
- b. 36.4 kJ/mol
- c. 50.2 kJ/mol
- d. 40.0 kJ/mol

20. The balanced equation for the reaction of nitrogen dioxide and fluorine is: $2 \text{NO}_2 + \text{F}_2 \rightarrow 2 \text{NO}_2\text{F}$. The proposed mechanism is:

Step 1: $\text{NO}_2 + \text{F}_2 \rightarrow \text{NO}_2\text{F} + \text{F}$ (slow)

Step 2: $\text{F} + \text{NO}_2 \rightarrow \text{NO}_2\text{F}$ (fast)

Which of the following are correct?

I. The mechanism supports an experimentally determined rate law of $\text{rate} = k[\text{NO}_2]^2[\text{F}_2]$

II. F is an intermediate

III. The reaction is first order in F_2

- a. I and II only
- b. I only
- c. I and III only
- d. I, II, and III
- e. II and III only

Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
B	A	A	D	A	C	C	B	B	E